

WASP-1b

R-2186

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-1_111205 · created 2026-08-13T23:22:46+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2455900.7587 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.128 +/- 0.037
Transit depth (Rp/Rs)^2	1.63 +/- 0.93 [%]
Orbital Inclination (inc)	88.73 +/- 1.4
Airmass coefficient 1 (a1)	1.051 +/- 0.014
Airmass coefficient 2 (a2)	-0.0174 +/- 0.0047
Scatter in the residuals of the lightcurve fit is	1.31 %
Best Comparison Star	#2 - [191, 247]
Optimal Method	PSF photometry
Transit Duration (day)	0.1012 +/- 0.0048

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.128 ± 0.037 vs 0.1036 (deviation 0.66σ)
Duration fit vs published	0.1012 d vs 0.1567 d (deviation 11.56σ)
Mid-time O–C	8.9 min
Depth SNR	3.5
Residual scatter	1.31 %

Light curve

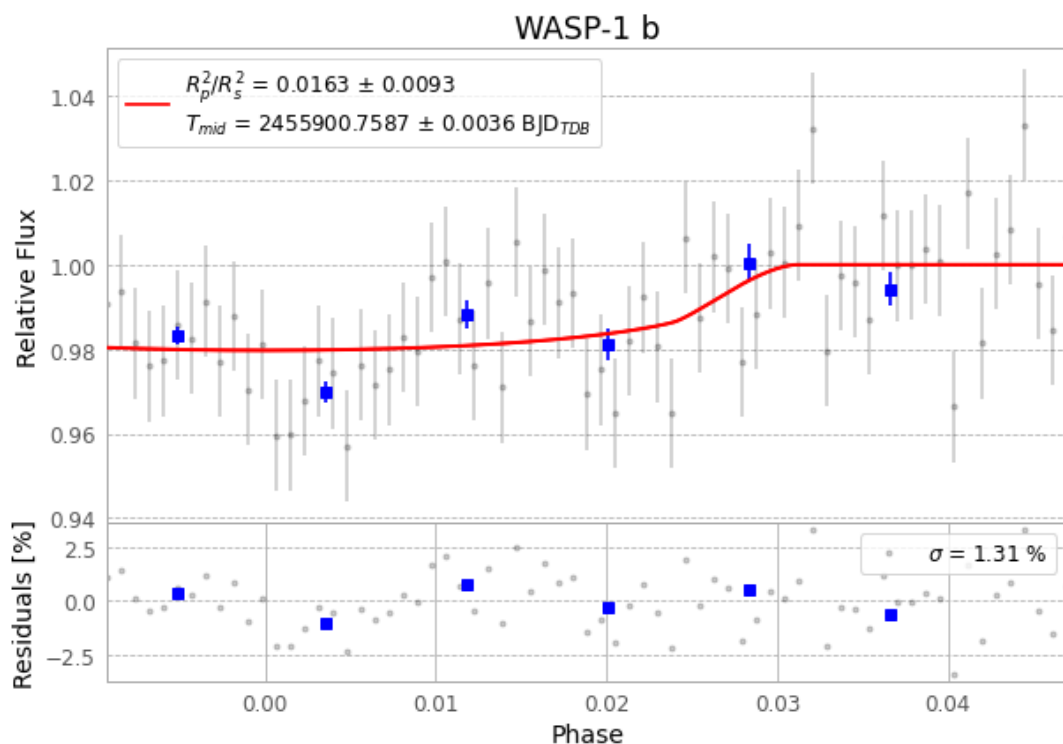


Figure 1 — FinalLightCurve_WASP-1 b_2011-12-04.png (SHA-256 3cfa25cd00ce1ba3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [185, 286], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dc57a9a22b2acf54...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

69 FITS frames (45 MB), WASP-1111205053313.FITS → WASP-1111205085712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-111205.log (d73f1fc72a2f...), FinalLightCurve_WASP-1 b_2011-12-04.png (3cfa25cd00ce...), FinalLightCurve_WASP-1 b_2011-12-04.csv (8540dff72c5...)

Compute resources

Wall time 112.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 23:22Z.